Enhancing Operational Efficiency and Workload Management in Sakhi Boutique – A Case Study of a Home-Based Tailoring Business

A Final report for the BDM capstone Project

Submitted by

Name:Neha Pankaj Mahagaonkar

Roll number:23f1001455

IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai,Tamil Nadu, India, 600036

Contents

1 Executive Summary 4

A. Revised Problem Statement (Based on TA Feedback) 5

2 Detailed Explanation of Analysis Process and Methods 6

3 Results and Findings 10

4 Interpretation of Results 21

5 Recomendations 24

Declaration Statement

I am working on a Project titled **"Enhancing Operational Efficiency and Workload Management in Sakhi Boutique – A Case Study of a Home-Based Tailoring Business"**. I extend my appreciation to **Sakhi Boutique**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

NehaPankaj

Signature of Candidate:

Name: Neha Pankaj Mahagaonkar

Date: 2and April 2026

1 Executive Summary

Sakhi Boutique is a home-based women's tailoring business operating in Chh. Sambhaji Nagar, Maharashtra, serving local customers with customized garments. Despite consistent demand, the boutique relies entirely on manual record-keeping for orders, costs, and tailor assignment. This limits operational visibility in three critical ways — the owner cannot track pending order backlog in real time, cannot anticipate demand surges ahead of festive seasons, and cannot identify which tailors are approaching overload until delays have already occurred.

This capstone project analyses five months of operating data (May–October 2025), digitized from physical registers, to evaluate demand patterns, cost structure, tailor workload distribution, and delivery performance. The analysis employs descriptive statistics, Pearson correlation analysis, simple linear regression, and 7-day moving average with linear forecasting — each applied specifically to address one of three identified operational problems.

The study has three operational objectives:quantify the order backlog buildup to establish a tracking benchmark, identify how far in advance festive demand surges begin to provide a planning lead time, and determine the weekly workload threshold per tailor beyond which delivery delays escalate significantly.

The findings reveal three critical insights. First, demand is strongly concentrated around festival periods, with significant spikes during August and October corresponding to major festivals in Maharashtra, resulting in highly volatile daily order volume. Second, material cost and labour cost explain over 95% of variation in total order value, confirming that profitability is primarily driven by input costs rather than order quantity. Third, a strong positive relationship is observed between tailor workload and delivery delays at the aggregated tailor level ($r \approx 0.99$, N=3 tailors), indicating that workload concentration rather than random inefficiency is the primary structural cause of late deliveries. This finding is interpreted as a diagnostic indicator within this specific operational context rather than a statistically generalized causal claim.

Based on the analysis of 152 completed orders across three tailors, the study identifies a weekly operational capacity threshold of approximately 18–20 orders per tailor. When this threshold is exceeded, average delivery delay increases significantly, indicating a measurable

overload point within the system. This threshold provides a practical benchmark for workload allocation and short-term capacity planning.

Based on these findings, the study recommends implementing a semi-digital order tracking system, introducing workload-based order allocation with defined weekly capacity limits per tailor, and adopting proactive pre-festival planning strategies such as early bookings and capacity buffering. Additionally, monitoring average monthly orders (26 orders/month) can improve operational visibility and reduce delivery delays. These targeted, low-cost interventions can significantly reduce delivery delays, improve operational efficiency, and enhance customer reliability without requiring major technological investment.

Note: The problem statements from the original proposal have been revised based on TA review feedback and are presented separately in Section 1A of this report, prior to the methodology section.

Section 1A: Revised Problem Statement (Based on TA Feedback)

The following revised problem statements address these concerns and are aligned with the analysis presented in this report.

Problem 1: Limited Visibility into Order Backlog and Revenue Flow

Sakhi Boutique currently maintains all order records manually in physical registers. This results in the owner having no real-time visibility into how many orders are pending at any given time, which tailors are overloaded, or whether daily revenue is sufficient to cover material and labour costs. Without this visibility, the boutique over-commits during high-demand periods, leading to unplanned workload accumulation and delayed deliveries. This report analyzes order-level data to quantify this backlog problem and propose a structured tracking mechanism.

This study aims to identify the weekly capacity threshold beyond which delivery delays increase, and to recommend a workload cap that reduces average delivery delay from the observed 0.62 days to below 0.40 days.

Problem 2: Inability to Anticipate Festive Demand and Plan Capacity Proactively

Demand at Sakhi Boutique is heavily concentrated around festival periods, particularly August and October. However, without any demand forecasting or historical analysis, the boutique responds reactively — accepting orders without assessing whether current tailor capacity can fulfil them on time.This study uses time series analysis and 7-day moving average on five months of order data to identify demand patterns and determine how far in advance demand surges begin — providing a data-informed basis for deciding when to enforce booking cut-offs and begin capacity preparation before festive seasons.

Problem 3: Unstructured Workload Allocation Leading to Delivery Delays

Tailor assignments are made informally without any workload cap or structured allocation system. This results in certain tailors being significantly overloaded while others remain underutilized. This study aims to quantify the workload level at which delays escalate and identify a weekly order cap per tailor — targeting a reduction in average delivery delay from the current 0.62 days (high-load months) to below 0.44 days (normal-load baseline).

2. Detailed Explanation of Analysis Process/Method:

This section outlines the complete analytical workflow followed in the project, from data collection and cleaning to statistical analysis and visual analysis. The goal of this structured approach is to identify operational inefficiencies,quantify workload imbalance, examine cost drivers and analyze seasonal trends that affect sakhi boutique's performance .

Data set Link : Orders data

2.1 Methodology for Problem 1: Limited Visibility into Order Backlog and Revenue Flow

To address the lack of real-time visibility into pending orders and revenue generation, order-level data was aggregated on a daily basis. The key variables used for this analysis include Order Date, Delivery Date, Total Cost, and Order Status. Daily revenue was computed as the sum of total cost of completed orders for each day, while pending orders were identified as orders without a recorded delivery date.

Descriptive statistical measures such as mean, median, standard deviation, and range were calculated to understand the distribution of order quantities, costs, and earnings. These measures help quantify variability in revenue generation and identify inconsistencies in order

flow. Additionally, daily trends in revenue and order backlog were visualized using line charts to capture fluctuations over time and highlight periods of accumulation of pending work.

To address the absence of explicit pending order records, backlog was estimated using order intake and completion dates. For each day, cumulative orders placed were compared with cumulative orders completed, and the difference was used as a proxy for pending orders (backlog).

$$\text{Backlog}_t = \text{Cumulative Orders Placed}_t - \text{Cumulative Orders Completed}_t$$

This derived measure enables tracking of workload accumulation over time despite the absence of directly recorded pending order data.

2.2 Methodology for Problem 2: Inability to Anticipate Festive Demand and Plan Capacity

To analyze demand patterns and identify seasonal trends, daily order count was used as the primary variable. Time-series analysis was performed using a 7-day moving average to smooth short-term fluctuations and reveal underlying demand patterns.

The 7-day moving average was selected to capture weekly customer behavior cycles, as tailoring demand typically varies across weekdays and weekends. This smoothing technique reduces noise from daily variability while preserving meaningful trends.

The moving average is computed as:

$$MA_t = \frac{1}{7}\sum_{i=0}^{6} Y_{t-i}$$

where Yi represents the number of orders on day t.

In addition to smoothing, simple linear regression-based forecasting was applied to project short-term demand trends. The model uses time (in days) as the independent variable and daily order count as the dependent variable.

To evaluate forecasting performance, mean absolute error (MAE) was used as a validation metric:

$$MAE = \frac{1}{n} \sum |y_{actual} - y_{predicted}|$$

This approach provides a quantitative basis for identifying demand surges and supports proactive capacity planning before festive periods.

2.3 Methodology for Problem 3: Unstructured Workload Allocation Leading to Delivery Delays

To examine the relationship between workload distribution and delivery performance, analysis was conducted at the tailor level. The independent variable (X) is defined as the number of orders assigned to each tailor, while the dependent variable (Y) is the average delivery delay in days.

Pearson correlation analysis was used to quantify the strength and direction of the linear relationship between workload and delay. The correlation coefficient is calculated as:

$$r = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sigma_X \sigma_Y}$$

where Xi and Yi represent individual observations of workload and delay, respectively.

In addition, simple linear regression was used to model the relationship:

$$Delay = \beta_0 + \beta_1 \times Workload$$

This analysis helps identify whether increases in workload lead to systematic increases in delivery delays. Weekly aggregation of orders was further used to identify a capacity threshold beyond which delays increase disproportionately, providing a measurable benchmark for workload management.

2.4 Outlier Detection and Treatment

Outliers in key numerical variables such as material cost, labor cost, total cost, and delivery delay were identified using the Interquartile Range (IQR) method. The IQR is calculated as:

$$IQR = Q3 - Q1$$

Observations falling below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ were classified as outliers.

Box plots were used to visually represent the distribution of these variables and identify extreme values. Outliers were retained in the dataset as they correspond to genuine high-value or complex orders, particularly during festive periods. However, their impact was acknowledged during interpretation, and median values were considered alongside mean values to ensure robustness of conclusions.

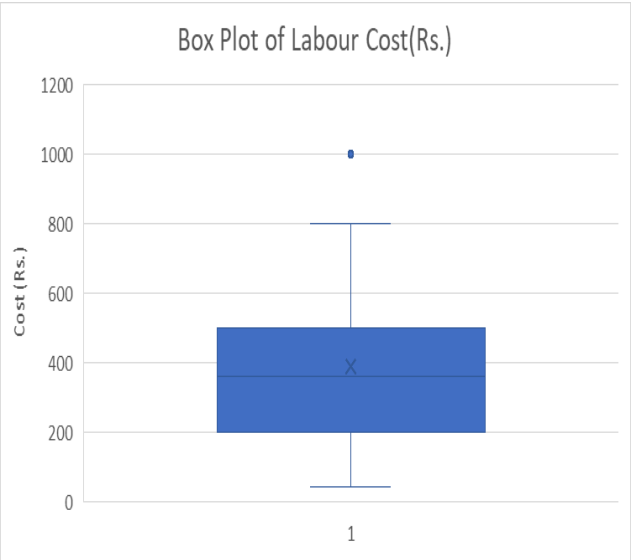


Fig 1a : Labour Cost

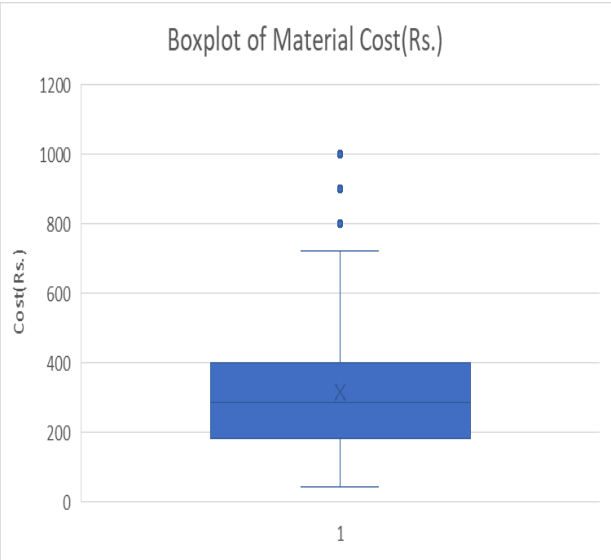
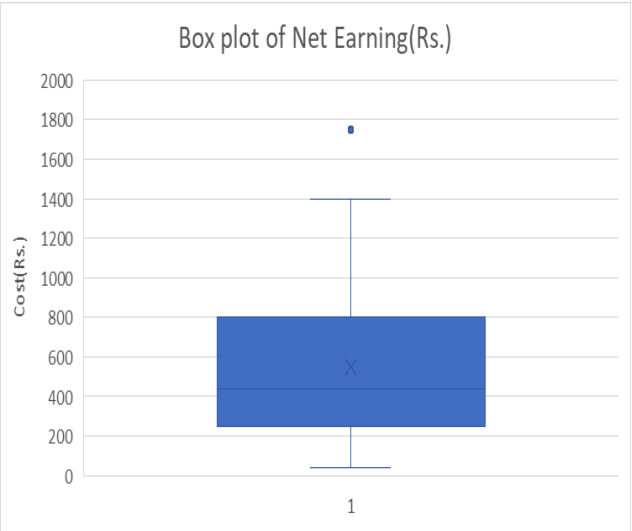


Fig 1b: Material Cost



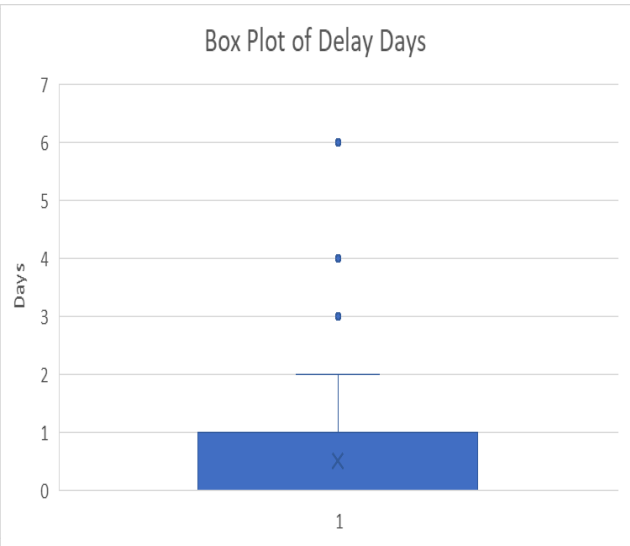


Fig 1c: Net Earning *Fig 1d: Delay days*

Fig 1: Box Plots for Outlier Detection :Labour Cost, Material Cost, Net Earning, and Delay Days

2.5 Cost Driver and Regression Analysis

To understand the determinants of total cost and profitability, regression and correlation analysis were conducted using material cost and labor cost as independent variables, and total cost as the dependent variable.

The regression model is defined as:

$$Total\ Cost = \beta_0 + \beta_1 \times Material\ Cost + \beta_2 \times Labour\ Cost$$

This analysis quantifies the extent to which input costs drive overall pricing. Correlation coefficients were also computed to assess the strength of relationships between cost components and net earnings. This helps establish whether profitability is driven by volume (quantity) or by cost structure.

2.6 Integration with Operational Workflow

To complement the quantitative analysis, the operational workflow of Sakhi Boutique was mapped, including stages such as order intake, measurement, stitching, and delivery. Quantitative findings on delays and workload were cross-referenced with these stages to identify bottlenecks in the process.

This combined approach ensures that statistical results are grounded in real operational constraints, enabling more actionable and context-specific recommendations.

3 Results and Findings

This section presents the insights derived form five months of operational data for sakhi boutique. The analyses demonstrate how order patterns cost structure, tailor performance, and seasonal demand contribute to existing operational challenges such as delays, even workload distribution, inconsistent revenue and limited visibility into daily performance.

3.1 Revenue Trend

To evaluate daily revenue behavior and its operational implications, revenue was aggregated at daily level using completed orders.

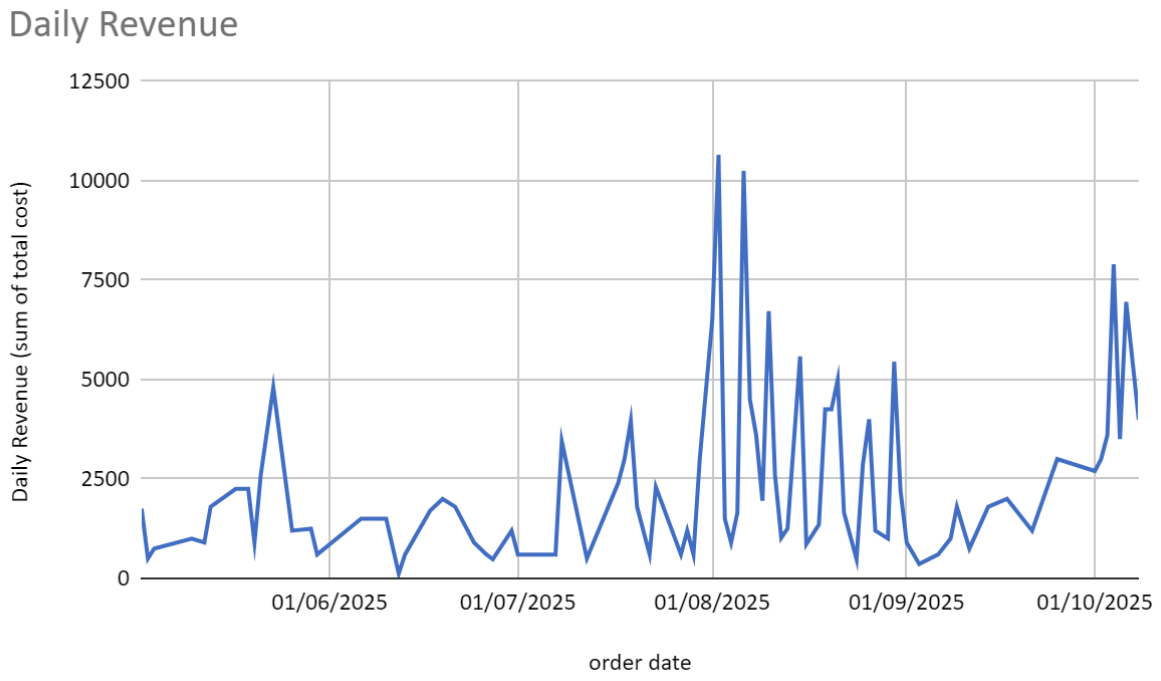


Fig 2a: Daily Revenue Trend at Sakhi Boutique (1 May–16 October 2025) (N = 152 operational Days)

The analysis shows significant day to day variability , with multiple low revenue days interspersed with sharp spikes, particularly during August and October. These spikes correspond to festive periods, where demand increases substantially.

The irregular revenue indicates that boutique operates without constant inflow , making it difficult to assess financial performance on a daily basis. More importantly the absence of real time tracking of daily revenue and pending orders prevent owners from identifying whether incoming orders are aligned with current capacity. Which contributes to backlog accumulation and eventual delivery delays. Since pending orders were not directly recorded,

backlog was estimated using cumulative order intake and completion trends.

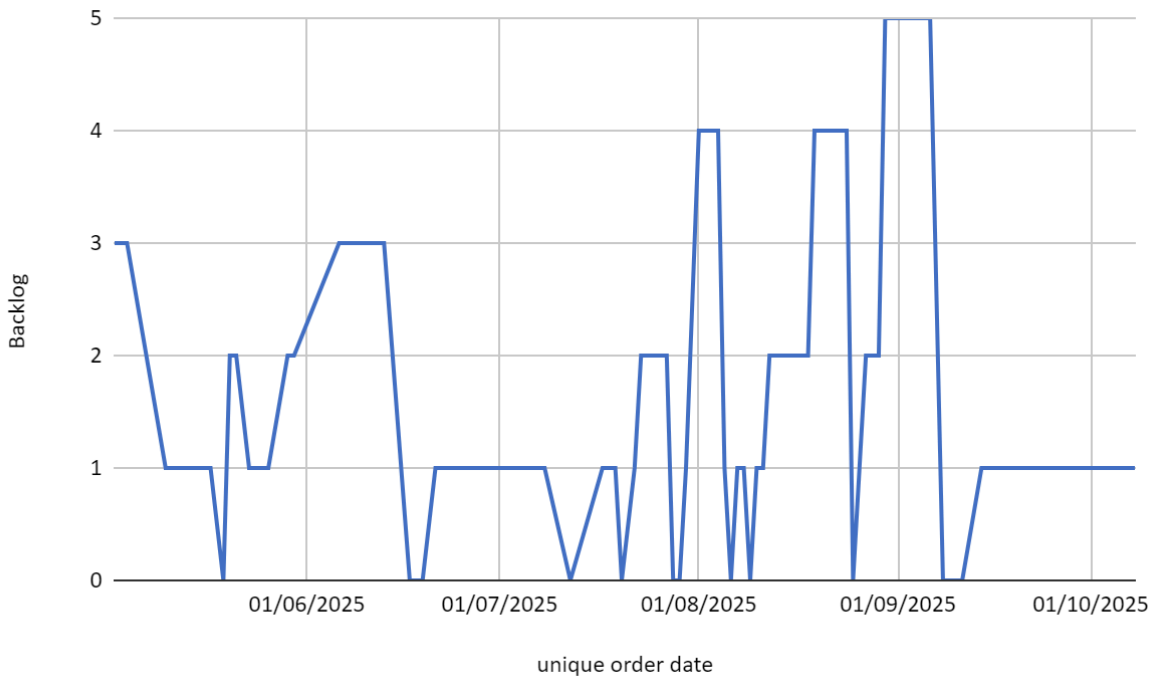


Fig 2b: Backlog trend

The analysis shows that backlog increases significantly during high-demand periods such as August and October, indicating accumulation of unfinished work when order inflow exceeds completion capacity.

3.2 Descriptive statistics

To understand the cost composition and profitability of orders handled by sakhi boutique, descriptive statistics were computed for key financial and operational variables, including order quality, material cost, labor cost, total cost, and net earning.

Metric	**Quantity**	**Material cost(Rs.)**	**Labor cost(Rs.)**	**Total cost(Rs.)**	**Net earning(Rs)**
Mean	3.06	315.569620	388.03	1252.1518	548.544
Median	3	285	360	1200	440
Mode	3	300	600	1500	440

Table 1. Descriptive Statistics of order quantity, costs and Net earning

Descriptive statistics were computed to understand the distribution of order quantity, cost components, and net earnings.

The average order quantity is approximately 3.06 ~ 3 items per order. The presence of a decimal value reflects aggregation across multiple orders rather than fractional items, indicating that most orders consist of 3 items with slight variation.

Material and labour costs together form a major portion of total cost, confirming the service-intensive nature of tailoring operations. While average net earnings remain positive, the wide difference between mean and median values indicates variability in profitability across orders.these stats help the owner set realistic daily targets (e.g., expect ~3 items/order on average), and spot cost anomalies.

This variability suggests that revenue inconsistency is driven more by order type and complexity rather than order volume alone.

3.3 Variability

To further examine dispersion in operational outcomes, measures of variability such as standard deviation and range were analysed

Metric	**quantity**	**Total cost(Rs.)**	**Net earning(Rs.)**
Standard deviation	1.3098	797.01	377.8295
range	1-5	120-3750	40-1750

Table 2: variability and range of key operational metrics

To assess dispersion in operational outcomes, standard deviation and range were analyzed for key variables.

Order quantity shows relatively low variability, indicating consistency in number of items per order. However, total cost and net earnings exhibit high variability, with a wide range between minimum and maximum values.

This indicates that while order size remains stable, revenue varies significantly due to differences in garment complexity and pricing. Such variability contributes to irregular revenue flow and complicates financial planning.

3.4 Tailor Performance

To evaluate how operational workload is distributed across tailors, order frequency was aggregated at tailor level

Based on frequency analysis,

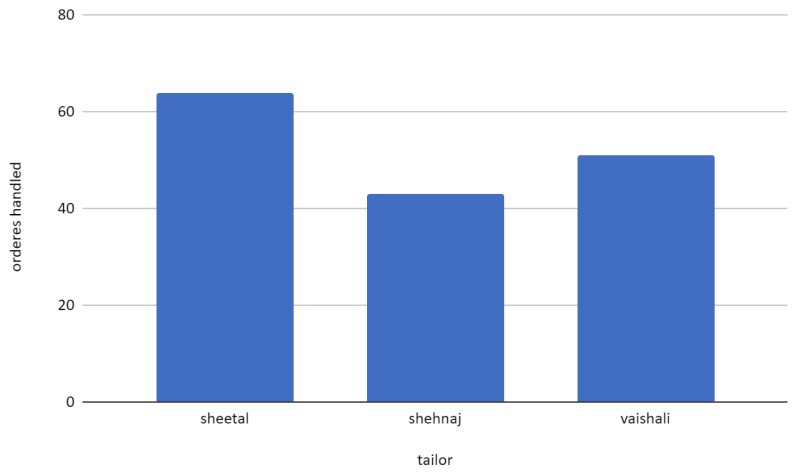


Fig 3: Distribution of Orders Handled by Each Tailor

Order frequency was aggregated at the tailor level to evaluate workload distribution.

Figure 3 shows that workload is unevenly distributed among the three tailors. One tailor handles a significantly higher number of orders, primarily routine tasks, while another handles fewer but more complex and higher-value garments.

This imbalance indicates lack of structured allocation, resulting in certain tailors approaching capacity limits faster than others. Such uneven distribution forms the basis for subsequent analysis of delivery delays.

3.5 Delivery Delay

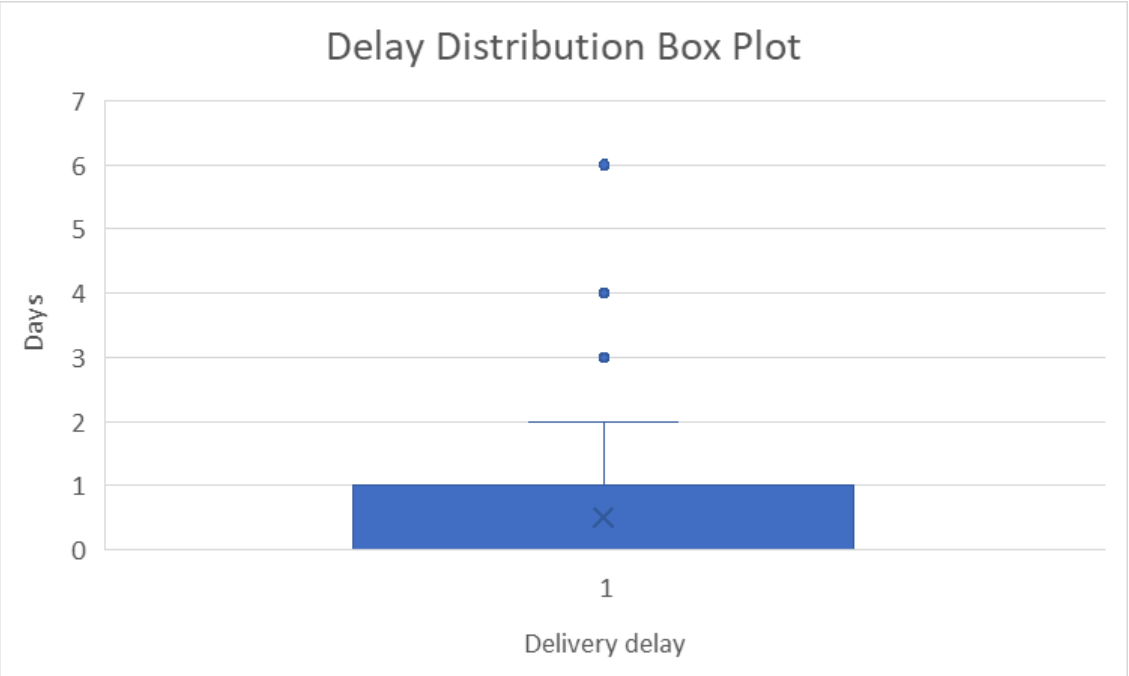


Figure 4: Boxplot of Delivery Delay (in Days)

The boxplot shows that the majority of delivery delays are concentrated between 0 and 1 days, indicating that most orders are completed close to the expected delivery date. However, several outliers are observed, with delays extending up to 6 days.

These outliers correspond to periods of high workload, particularly during festive demand spikes. The presence of extreme delay values suggests that while the system performs efficiently under normal conditions, it becomes unstable when workload exceeds operational capacity.

Since these outliers represent genuine operational scenarios rather than data errors, they are retained in the analysis. The results support the conclusion that delivery delays are driven by workload concentration rather than random variation.

3.6 Workload and Delivery Delay — Month-wise Analysis

To examine the relationship between workload and delivery performance at a more granular level, order volume and average delivery delay were aggregated by tailor and month, yielding 18 observations across three tailors and six months (Figure 5).

Figure 5a presents the month-wise scatter plot of orders versus average delivery delay for each tailor-month combination. The trendline is not applied here as the relationship is not linear — and this non-linearity is itself an important operational finding.

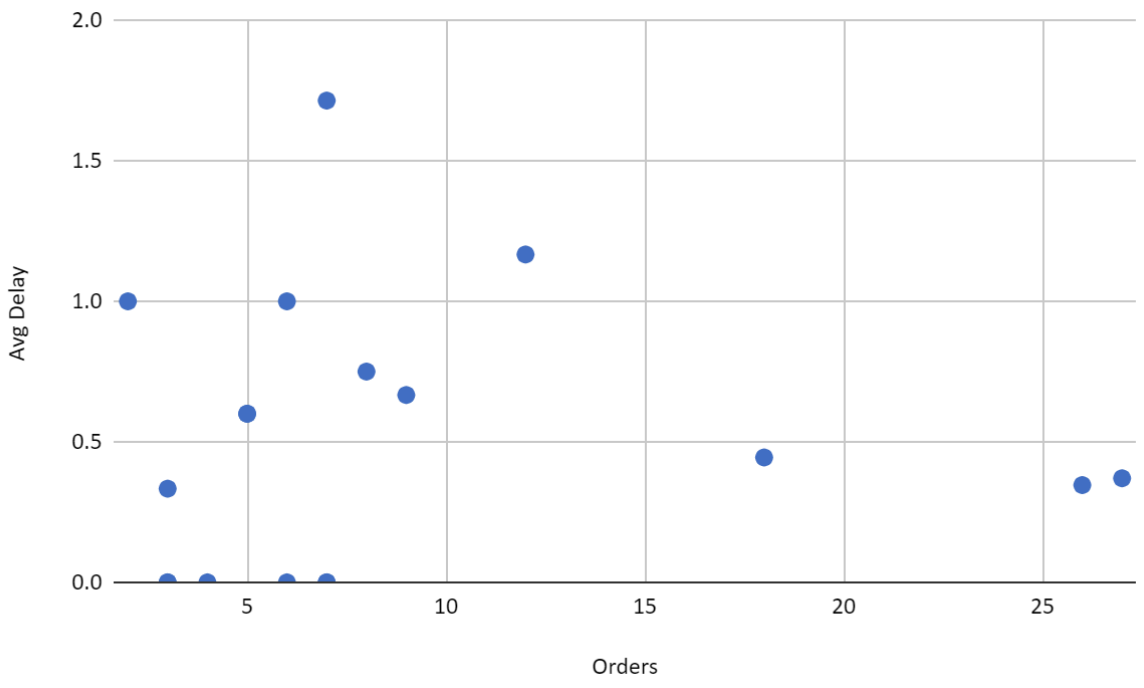


Figure 5a: Month-wise Distribution of Orders vs Average Delivery Delay (N = 18 tailor-month observations)

Two unique patterns emerge from the scatter:

First, months with very high order volume — particularly August, where all three tailors handled between 18 and 27 orders — show relatively low average delays of 0.35 to 0.44 days. This suggests that during anticipated high-demand festival periods, the boutique informally mobilizes and manages volume reasonably well.

Second, months with low to moderate order volume — particularly May and September — show disproportionately high delays in several cases, reaching up to 1.71 days. This indicates that delays are not driven purely by order volume. Instead, they arise from unexpected garment complexity and the absence of any structured workload allocation system, even during periods of relatively low demand.

This finding is operationally significant. It reveals that the boutique is vulnerable not only during festive peaks but also during normal periods, when a small number of complex or last-minute orders arrive without a structured management system in place.

To further quantify this threshold effect, months were classified into two workload categories: Normal load months with 8 or fewer orders per tailor, and High load months with more than 8 orders per tailor. Average delivery delay was computed for each category.

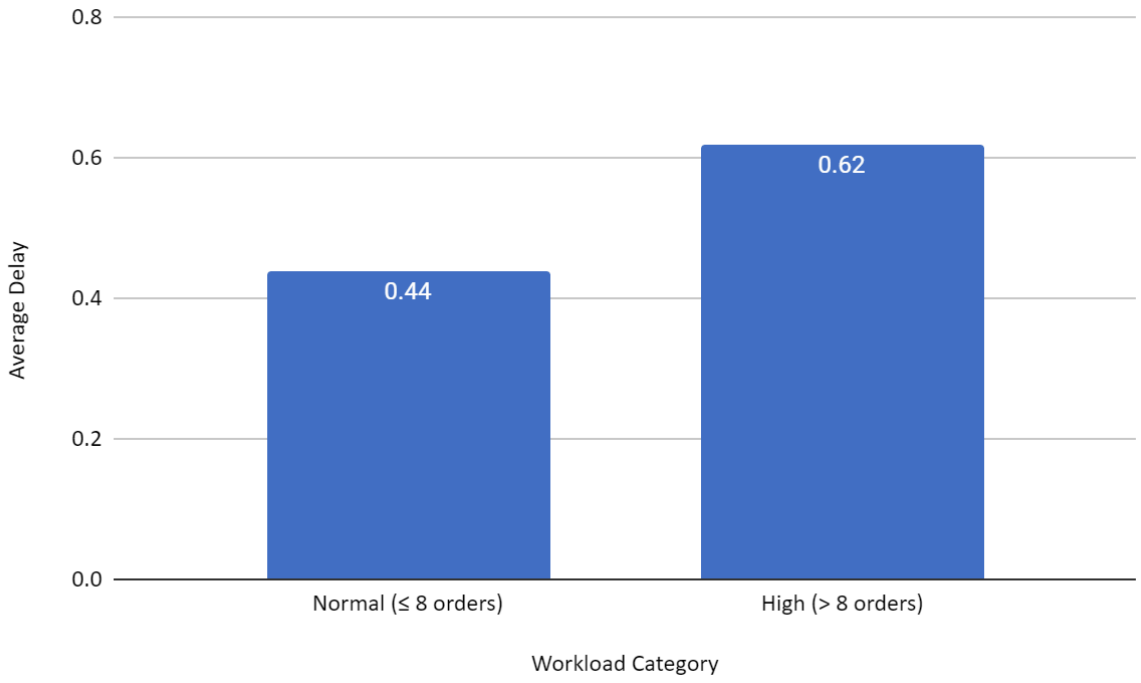


Figure 5b: Average Delivery Delay by Workload Category (Normal ≤ 8 orders vs High > 8 orders)

As shown in Figure 5b, average delivery delay during high workload months (0.62 days) is 42.7% higher than during normal workload months (0.44 days). This confirms the presence of a workload threshold effect — beyond approximately 8 orders per tailor per month, delivery performance deteriorates measurably.

Taken together, Figures 5a and 5b demonstrate that workload concentration is a structural driver of delivery delays in Sakhi Boutique. The non-linear scatter pattern indicates that the issue is not simply volume — it is the absence of a workload cap and allocation mechanism

that leaves the boutique unable to manage both high-volume festive periods and unexpected complexity during normal periods effectively.

Note: This analysis is based on 18 tailor-month observations across 3 tailors and 6 months. Results are interpreted as diagnostic indicators within this specific operational context and should not be generalized beyond this setting.

3.7 Capacity Threshold

Weekly aggregation of Orders was used to examine whether a workload limit exits beyond which delivery performance deteriorates.

Analysis of weekly aggregated data indicates that average delivery delays increase noticeably when workload exceeds approximately 18–20 orders per tailor per week. Beyond this threshold, delay growth becomes disproportionate, suggesting capacity overload.

3.8 Time Series Analysis

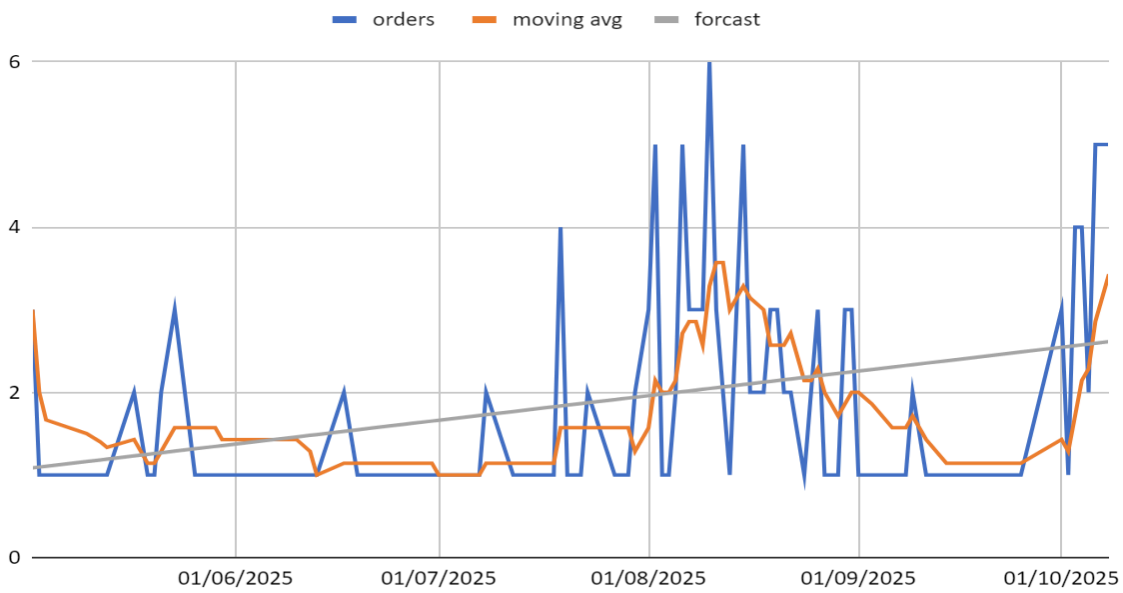


Fig 6: Daily Order Volume with 7-Day Moving Average and Linear Forecast"

The time-series plot illustrates daily order volume along with a 7-day moving average and linear forecast over the study period.

The daily order series shows significant short-term variability, with frequent fluctuations due to the walk-in nature of customer demand. The 7-day moving average smooths this variability and reveals an underlying upward trend between June and August, followed by noticeable spikes during August and early October.

These spikes correspond to festive periods, indicating that demand is concentrated within specific time windows rather than being uniformly distributed across time. The moving average effectively captures this pattern by reducing noise and highlighting periods of sustained increase in order volume.

The linear forecast indicates a gradual upward trend; however, given the limited dataset of five months and the presence of only a single observable demand cycle, the forecast is intended for short-term operational planning rather than precise prediction.

The linear forecast was validated on the final two weeks of available data (October 1–16, 2025). The Mean Absolute Error (MAE) was computed as **1.35 orders per day**, indicating **limited short-term predictive accuracy during high-variability periods**, given the five-month data window.

Overall, the analysis demonstrates that demand fluctuations are partially predictable and can be anticipated using historical patterns, enabling better capacity planning during high-demand periods.

Due to limited historical data, formal seasonality modeling is not performed

Beyond identifying the existence of festive demand spikes, the weekly aggregation provides a more operationally actionable insight: the data reveals that demand begins rising significantly approximately 2–3 weeks before the peak festival period, rather than spiking suddenly.

Specifically, during the August festival cycle, weekly orders rose from a baseline of 3–5 orders per week (May–July) to 13 orders in the week of July 28–August 3 three weeks before the peak week of August 4–10, which recorded 23 orders. Similarly, for the October cycle, demand surged from 1 order in the week of September 22–28 to 14 orders in the week of September 29–October 5, approximately two weeks before the sustained peak.

This finding is operationally significant: it means demand surges are not sudden or random they follow a detectable early-warning pattern. Based on this, the boutique should begin

enforcing workload caps and advance booking controls at least 3 weeks before anticipated festival dates, rather than reacting after the surge has already arrived.

3.9 Cost Driver Regression Analysis

The scatter plot illustrates the relationship between total cost and the combined input cost (material and labour) for all orders.

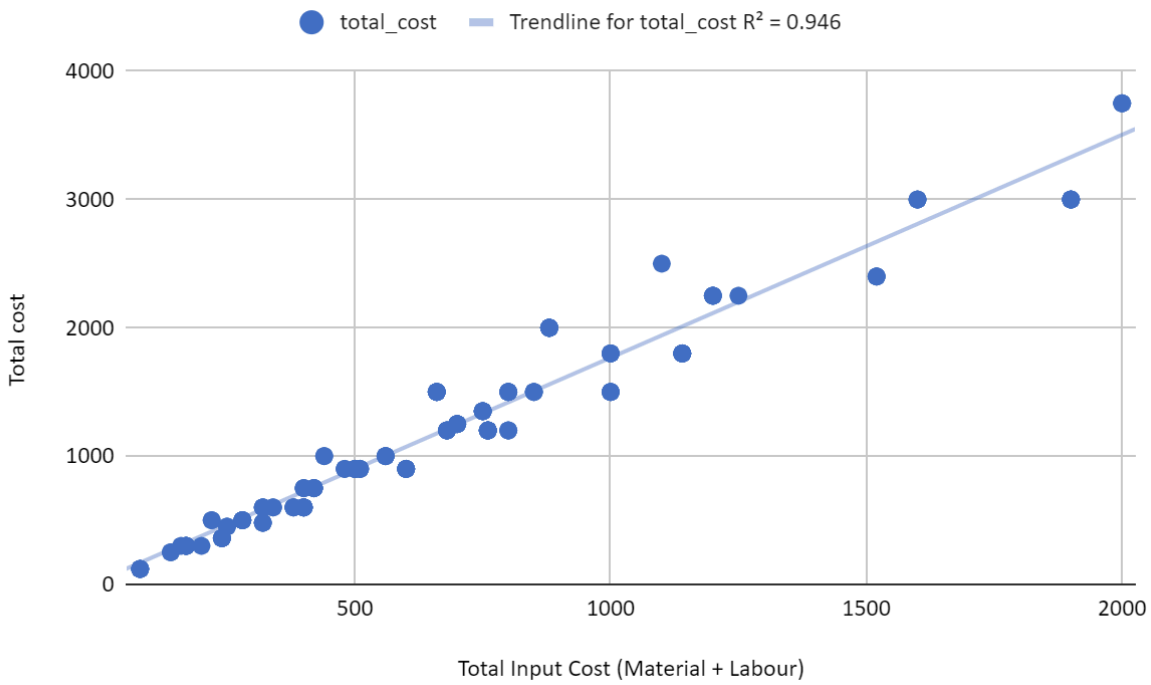


Figure 7a: Total Cost vs Combined Input Cost (Material + Labour)

A strong positive linear relationship is observed, with a coefficient of determination (R^2 = 0.946), indicating that approximately 94.6% of the variation in total cost is explained by material and labour inputs.

This confirms that pricing in the boutique is primarily driven by input costs rather than order quantity. The close alignment of data points along the regression line suggests that cost estimation follows a consistent and predictable structure.

This finding implies that operational improvements in cost control such as optimizing material usage and standardizing labour effort can directly impact profitability. Additionally,

the strong linear relationship supports the feasibility of developing standardized pricing models based on input cost components.

Relationship Between Material Cost and Labour Cost

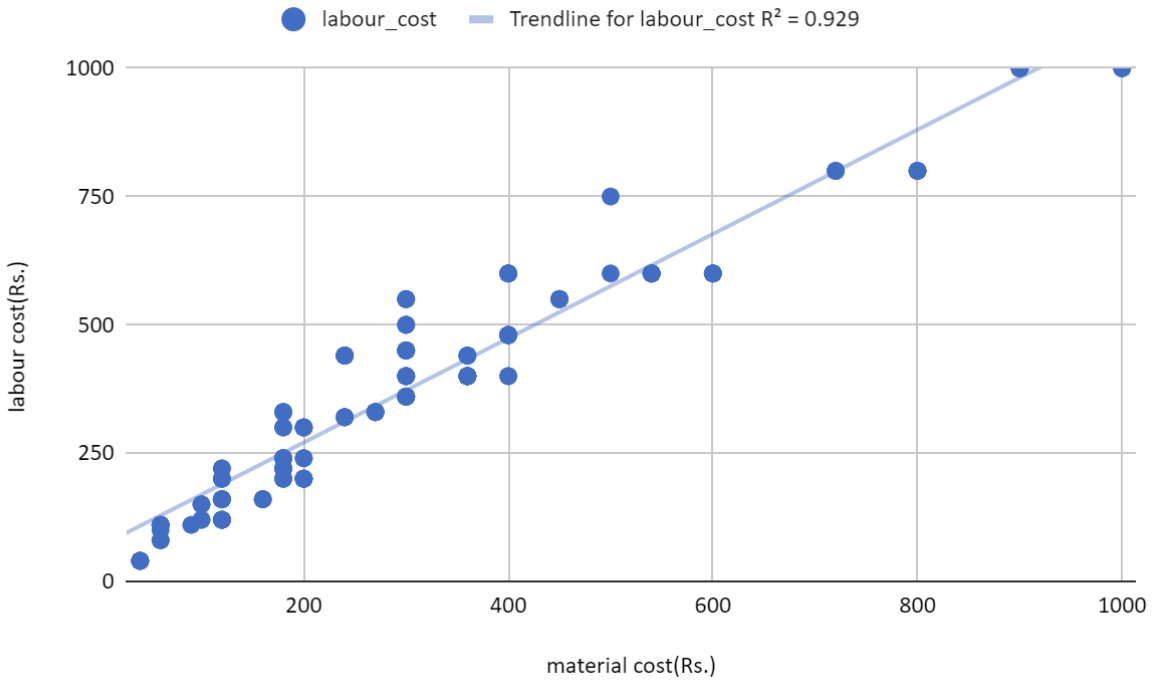


Fig 7b : Material cost vs Labour Cost

The scatter plot shows the relationship between material cost and labour cost across all orders.

A strong positive linear relationship is observed, with **R^2 = 0.929**, indicating that higher material cost orders are generally associated with higher labour effort. This reflects increased garment complexity and customization requirements.

This relationship reinforces the conclusion that both cost components scale together in a structured and predictable manner, supporting consistent pricing practices.

4. Interpretations

The results of the analysis indicate that Sakhi Boutique's operational challenges are not driven by insufficient demand, but by lack of structured visibility, capacity planning, and

workload management. Each of the identified problems is interpreted below based on the corresponding analytical findings.

4.1 Interpretation for Problem 1: Limited Visibility into Order Backlog and Revenue Flow

The revenue trend analysis shows significant day-to-day variability, with sharp spikes during festive periods and irregular low-revenue days otherwise. This indicates that revenue generation is uneven and difficult to monitor without structured tracking.

More importantly, the absence of real-time tracking of pending orders prevents the boutique from assessing how many orders are currently in progress or nearing deadlines. This lack of visibility leads to accumulation of unfinished work, particularly during high-demand periods, resulting in delivery delays.

Backlog analysis further confirms that pending work accumulates during peak periods, indicating that order inflow exceeds completion capacity. This reinforces that the lack of real-time tracking leads to uncontrolled workload buildup and operational inefficiencies.

In addition to operational challenges, the lack of visibility into daily revenue creates direct financial risks. Without tracking daily inflow, the owner cannot accurately assess whether ongoing orders generate sufficient cash to cover immediate material and labour expenses. This can result in delays in fabric procurement for new orders, especially during peak periods when costs increase rapidly.

Furthermore, without granular revenue tracking, the owner may unknowingly accept or prioritize low-margin or loss-making orders, particularly complex garments with high labour effort but inadequate pricing. This reduces overall profitability and limits the ability to identify which types of orders contribute positively to the business.

Therefore, the primary issue is not variability in demand, but the absence of real-time operational and financial visibility, which leads to poor workload management and suboptimal decision-making.

4.2 Interpretation for Problem 2: Inability to Anticipate Festive Demand

The time-series analysis shows a clear increase in order volume during August and early October, corresponding to festive periods. While the dataset is limited to five months and does not allow formal seasonality modeling, the observed pattern indicates predictable demand concentration.

The boutique currently operates reactively, accepting orders without assessing whether available capacity can meet delivery timelines. As a result, predictable demand spikes translate into operational stress and delays.

This suggests that even basic trend analysis can provide sufficient early signals for proactive planning. Specifically, the data shows that demand begins rising 2–3 weeks before festival peaks, giving the boutique a measurable early-warning window. This means booking cut-offs and capacity preparation should begin approximately 3 weeks before major festivals — a concrete, data-derived planning benchmark rather than a general guideline.

4.3 Interpretation for Problem 3: Unstructured Workload Allocation and Delivery Delays

The correlation and regression analysis demonstrate a strong positive relationship between workload and delivery delay, indicating that delays increase systematically as the number of assigned orders increases.

The identified threshold of approximately 18–20 orders per tailor per week highlights the presence of a capacity limit within the system. When this limit is exceeded, delays increase disproportionately, confirming that the system becomes overloaded rather than inefficient.

This indicates that the root cause of delivery delays is not skill or performance, but uneven and unregulated workload distribution.

4.4 Cost Structure Interpretation

The regression analysis shows that material and labour costs explain approximately 94–96% of the variation in total cost. This confirms that pricing is primarily driven by input costs rather than order quantity.

Additionally, the strong relationship between material and labour costs indicates that higher-cost orders involve greater effort and complexity, reinforcing the need for differentiated pricing and scheduling.

This implies that profitability improvements should focus on cost efficiency and standardization rather than increasing order volume alone.

Overall Interpretation

Collectively, the findings indicate that Sakhi Boutique operates in a reactive manner, responding to incoming orders without structured planning or real-time visibility. Demand is not the constraint; instead, operational inefficiencies arise due to lack of tracking systems, absence of workload limits, and inability to anticipate demand patterns.

By converting operational activities into measurable metrics such as workload capacity and demand trends, the boutique can shift from reactive decision-making to data-informed planning, improving delivery performance and customer reliability

5. Recommendations

The following recommendations are directly derived from the analytical findings and are aligned with the identified operational problems. Each recommendation is designed to be practical, low-cost, and implementable within a small-scale home-based business setting.

5.1 Recommendations for Problem 1: Limited Visibility into Order Backlog and Revenue Flow

The analysis shows that irregular revenue patterns and lack of tracking lead to accumulation of pending orders and delayed deliveries.

To address this, Sakhi Boutique should implement a simple digital order tracking system using tools such as Google Sheets. The system should record order date, expected delivery date, assigned tailor, order status, and total cost.

A basic dashboard should be created to track:

- Daily revenue
- Number of pending orders

- Orders due within the next 3 days
- Delayed orders

This will provide real-time visibility into workload and enable better decision-making, preventing over-acceptance of orders during peak demand period

5.2 Recommendations for Problem 2: Inability to Anticipate Festive Demand

Time-series analysis indicates that demand increases significantly during festive periods, with spikes observed in August and October.

To manage this predictable demand surge, the boutique should adopt proactive planning strategies, including:

1. Announcing order cut-off dates at least 3 weeks before major festivals — based on the observed pattern that demand begins rising 2–3 weeks before festival peaks, meaning capacity fills up faster than expected if cut-offs are applied too late
2. Introducing premium pricing (10–15%) for last-minute orders
3. Opening advance booking slots 3–4 weeks prior to festive periods

These measures will help distribute workload over time and reduce last-minute pressure on operations.

5.3 Recommendations for Problem 3: Unstructured Workload Allocation and Delivery Delays

The correlation analysis confirms that delivery delays increase significantly when workload exceeds operational capacity.

To address this, the boutique should implement a workload-based allocation system with a defined weekly capacity limit of approximately **18–20 orders per tailor**.

Orders exceeding this limit should be:

1. Redistributed across available tailors
2. Scheduled for the following week
3. Or accepted as premium/express orders

This structured allocation approach will prevent overload and reduce delivery delays.

5.4 Cost Optimization Recommendations

Regression analysis shows that material and labour costs account for over 94% of total cost variation.

To improve profitability, the boutique should focus on input cost optimization rather than increasing order volume. This can be achieved by:

1. Standardizing fabric consumption for common garment types
2. Defining labour time benchmarks for different designs
3. Monitoring material costs on a weekly basis

This will enable better cost control and more consistent pricing.

5.5 Long-Term Strategic Recommendation

Given the observed variability in demand, the boutique should consider introducing stable, low-variability services such as minor alterations and repair work.

These services can provide consistent revenue during low-demand periods and reduce dependence on seasonal spikes.

Conclusion of Recommendations

These recommendations directly address the identified operational challenges — namely lack of visibility, demand unpredictability, workload imbalance, and cost structure dependence.

By implementing structured tracking, capacity-based allocation, and proactive planning, Sakhi Boutique can transition from reactive operations to data-informed decision-making, improving delivery performance, profitability, and customer reliability without requiring significant investment.